OMAR EZ-ELDIN ABDULLAH

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PROFILE

AI Engineer and Computer Science graduate from Cairo University (FCAI) specializing in Agentic AI and Applied Deep Learning. Experienced in engineering end-to-end RAG architectures, local-first LLM agents, and automated computer vision pipelines. Proficient in PyTorch and LangGraph, with a strong focus on data engineering, domain adaptation, and deploying optimized local inference systems for highly accurate, low-latency AI applications.

EDUCATION

B.Sc in Computer Science **10/2022 – 07/2026**

Faculty of Computers and Artificial Intelligence, Cairo University *Cairo, Egypt*

- **GPA:** 3.3/4.0

WORK EXPERIENCE

Software Engineering Intern **07/2025 – 10/2025**

Microsystems Cloud Business Solution (MCBS) *Cairo, Egypt*

- **Frontend & Web Development:** Developed modular web applications and integration workflows using TypeScript, React, and Angular, managing build pipelines (Vite, Webpack) and E2E testing to ensure system reliability.
- **Network & Endpoint Security:** Conducted system security assessments by identifying open ports, discovering endpoint vulnerabilities, and executing penetration testing to document mitigation strategies.

AI Trainer / LLM Evaluator **05/2024 – 01/2026**

Scale AI *San Francisco, CA (remote)*

- Evaluated and refined 250+ LLM outputs across code review, multilingual Q&A, image evaluation, and math.
- Authored detailed justifications, collaborating with QMs to resolve edge cases and enforce strict quality standards.

PROJECTS

Seed Bank (Grad Project) Seed Quality Classification **10/2025 – 07/2026**

PyTorch, YOLOv8, Faster R-CNN, EfficientNet-B2, U2-Net, Synthetic Data Generation

An automated computer vision platform that detects, counts, and classifies individual agricultural seed defects to streamline large-scale quality control. It includes a custom synthetic data engine to overcome real-world data scarcity, ensuring highly reliable performance in unpredictable environmental conditions.

- **Dual-Pipeline Architecture:** Engineered a YOLOv8 single-stage pipeline for real-time inference and a modified Faster R-CNN + EfficientNet-B2 two-stage pipeline for high-accuracy defect classification, achieving **97.47% accuracy** and **96.89% F1-score** on maize defect classification.
- **Synthetic Data Engine:** Built "MultiSeedGen" to apply domain randomization (clutter, occlusion, shadows, camera noise) and close the synthetic-to-real gap, alongside integrating U2-Net segmentation to remove background noise and isolate seed contours.
- **Production Workflow:** Established a model registry and benchmarking workflow to version, evaluate, and archive candidate models against fixed baselines before production promotion.

Agentic AI Research Assistant

LangGraph, Ollama, Qwen3.5-4B, ChromaDB, BM25, BGE Embeddings, Sentence-Transformers, Streamlit, Tavily

A local-first, autonomous AI research assistant that analyzes specialized digital libraries to provide accurate, fully cited answers. The agent autonomously decides when to use local knowledge vs. web search, evaluates the quality of its own retrieved evidence, and dynamically adjusts its research strategy to ensure comprehensive, hallucination-free responses.

- **Agentic Orchestration (LangGraph):** Architected a multi-step, stateful agent workflow featuring "Fast" (low-latency) and "Deep" (high-recall) execution profiles, utilizing bounded conversational memory (last 4 turns) to resolve complex follow-up queries autonomously.

- **Self-Correction & Dynamic Strategy:** Engineered autonomous reasoning loops where the agent grades the relevance of retrieved local passages. If evidence is deemed insufficient, the agent dynamically deploys fallback retrieval strategies (step-back prompting, fusion, and specific query rewriting) before generating a final answer.
- **Tool Integration & Hybrid Retrieval:** Implemented a hybrid search tool (BM25 + ChromaDB vector search) alongside the Tavily Web Search API, enabling the agent to seamlessly route queries between offline private data and live internet data based on real-time tool evaluation.

Voice Gender Classification Microservice
Wav2Vec 2.0, *Scikit-learn*, *FastAPI*, *MLflow*, *Docker*, *AWS EC2*

An end-to-end audio processing system that accurately predicts voice gender directly from raw audio files. The predictive model is deployed as a fast, scalable web service capable of processing diverse audio formats.

- **Model Architecture:** Built a speech classification pipeline using a frozen Wav2Vec 2.0 backbone (Layer 6) and Logistic Regression, achieving **100% in-distribution** validation accuracy and **99.2% out-of-distribution (OOD)** accuracy on the Google FLEURS dataset.
- **Audio Preprocessing:** Engineered an in-memory preprocessing pipeline (16kHz resampling, mono conversion, duration checks) with format fallbacks to reliably decode diverse audio files (16/24/32-bit WAV).
- **Deployment & MLOps:** Built an asynchronous FastAPI backend, tracked training experiments in MLflow, and containerized with Docker for reproducible AWS EC2 deployment.

Advanced RAG Architectures
LangChain, *ChromaDB*, *Hugging Face* (*BGE v*1.5), *LangChain-Groq*.

A deep-dive into advanced Retrieval-Augmented Generation (RAG) strategies designed to enable AI systems to deeply understand complex user questions and translate them into highly accurate database queries.

- **Query Processing:** Implemented advanced query translation and routing strategies including Multi-query, RAG-Fusion, Step-back prompting, HyDE, and Text-to-SQL/Cypher generation.
- **Indexing & Representation:** Engineered chunk optimization, multi-representation indexing (RAPTOR) to capture rich semantic meaning across diverse data formats.
- **Retrieval Refinement:** Integrated advanced retrieval refinement techniques including Re-ranking, Corrective RAG (CRAG), and Self-RAG to dynamically filter, rank, and refine information before generation.

Rock Loyalty System
Angular 18, *TypeScript*, *RxJS*, *JWT*, *OAuth*2

- Developed a secure, multi-tenant customer portal allowing restaurant patrons to manage loyalty points and restaurant admins to seamlessly oversee daily operations.
- Implemented robust user authentication and access control features, ensuring highly secure logins and seamless, uninterrupted session management across the entire platform.

SKILLS

Machine Learning & Computer Vision:
PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, ONNX, rembg U²-Net, Faster R-CNN, Wav2Vec2, Roboflow Inference.

Data Science & Analysis:
NumPy, Pandas, Matplotlib, Seaborn

Programming Languages:
Python, C++, SQL, Java, Dart, JavaScript/TypeScript.

LLM & MLOps:
LangChain, LangGraph, LangSmith, ChromaDB, Sentence Transformers, BM25 Hybrid Retrieval, Hugging Face, Ollama, llama.cpp, MLflow, Docker, AWS.

Backend & Core Engineering
System Design, REST APIs, PostgreSQL, Object-Oriented Programming (OOP), Data Structures & Algorithms.

CERTIFICATES

2023 ECPC Qualifications Day 5 , MCBS internship completion , Digitera Certificate

LANGUAGES

Arabic
Native

English
Proficient